Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 6 to Draft 7

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Scope

Draft 7 responds to ChatGPT's referee report on Draft 6 (seven recommendations, all adopted), adds the material on representation levels requested by David Joyner, and, following his decision to stop the research at level two, adds a Future Directions section (new Section 15) stating what a general result would need. No new theorem.

Decision to stop at level two

Recorded in Section 15.1. The reasons: levels are unbounded, so no finite list proves Target B; the level-two proof uses three features that all fail at level three (class-two group with derived subgroup of order 3; $\Theta_\rho \circ r$ supported on a central subgroup; a central subgroup $\hat{A}$ large enough to detect every $t \notin Q_3$ through central characters); and the referee's analysis agrees that the central problem is now an inductive mechanism, not a computation. Target B is therefore stated as: proved for every row of level ≤ 2, open beyond.

The referee's seven corrections

1. **Section 13.5 (now "Beyond level two").** The false statement that Q_4/Q_{3m+1} is abelian is removed ($[1+\varpi^2 i, 1+\varpi^2 j]$ has depth 9, so Q_4/Q_{10} is nonabelian); the candidate abelian normal subgroup is $Q_{\lceil (3m+1)/2 \rceil}/Q_{3m+1}$, i.e. Q_5/Q_{10} at $m=3$. The sentence on linear χ now refers to D_8-invariant sums. The "single free orbit" statement is corrected: the 243 genuine characters of $\hat{H}_1 = \tilde{V}$ form nine free orbits, of which Λ is one. H_m and $\hat{H}_m$ are standardized with the cutoff $N_m = \max(3m+1, 6)$. The two statements (B1), (B2) are formulated separately, as recommended, with the remark that a failure of (B1) would be a genuine obstruction while a failure of universal freeness would not. The referee's explicit non-integrality example ($\chi(a+bi) = \zeta^a$, $h = (0,0,0,1,1)$, inner product $1/3$) was reproduced from the tables and is quoted.

2. **Howe-duality remark (now Remark 16.1).** The pointwise kernel K_θ diverges ($K_\theta(1,1) = 27 \sum \dim \rho^2$); the remark now works with the finite truncations $K_\theta^{(m)}$ over the irreducible representations of H/K^{m+1} with central character θ, which are honest class functions on $\tilde{H} \times H/K^{m+1}$ with the explicit formula, and Target B is restated as "$K_\theta^{(m)}$ is a character for every m". The Schrödinger-model language in Section 2.3 is now explicitly an analogy: $\tilde{U}_3/N_6$ is abelian

and $\tilde{U}_1/N_6$ is not a Heisenberg group with nondegenerate symplectic quotient; what is used is Clifford theory for an abelian normal subgroup.

3. **Self-contained level-two theorem.** New Lemma 13.7 (the group $H = Q_3/Q_7$): $H' = Q_6/Q_7$, $Z(H) = Q_4/Q_7$, $243 + 54$ characters, and $3\sum_{\chi'(1)=3}\chi' = \mathrm{Reg}_H - \mathrm{Inf}\,\mathrm{Reg}_{H/H'}$ together with the ω-isotypic form, proved by hand; Step 2 of the theorem now cites it.

4. $\beta \neq 0$ **made explicit** (Lemma 10.2(b)). For $E = F(j)$: $(1 + aj)^3 = 1 + a^3\varpi j + 3a^2\varpi + 3aj$ with $v_E(3) = 4$, so cubes of $U_E^{(1)}$ have leading term at E-depth 3 and their E-depth-5 component is determined; hence $E^\times/F^\times E^{\times 3}$ (order 9) has basis the classes of $1 + j$ and $1 + \varpi^2 j$, the commutator pairing is nondegenerate on it, and $[\widetilde{1+j}, \widetilde{1+\varpi^2 j}] = \kappa_{1,5}(1,1) = \zeta^{2\beta} \neq 1$. The consistency with the top-level Hilbert pairing of E at level $6 = pe_E/(p-1)$ is noted.

5. **Proof-status language.** "The last two steps are exact finite computations" is replaced by "all by hand; scripts 13–15 corroborate". Proposition 13.4 and Theorem 13.5 now carry the label [P^c] and begin "Assume Theorem 12.1", making the dependence on Target A explicit.

6. **Reproducibility.** Script 13 certifies the group law of Q_3/Q_7 deterministically (associativity on all 729^3 triples) and certifies the character table exactly (each rounded row is a common eigenvector of all class-sum matrices, computed in $\mathbb{Z}[\zeta_3]$ with integer arithmetic, and has the right norm); script 15 certifies that λ_0 is a character of $\hat{H}$ on all (generator, element) pairs, and its certificate now includes the multiplication table of Q_1/Q_3; a new `verify_certificate.py` re-reads the certificate and checks, without the cover, that the tables define an action of a group of order 27 with no fixed points and the listed orbits. No sampled checks remain in scripts 13–15. The scripts used for this draft are tagged `draft7-scripts` in the repository, and Appendix B says so.

7. **Prime degree.** "Every element of $D \setminus F$ generates a maximal subfield" is restricted to prime degree.

Material on levels (Section 2.1, new paragraph)

Level as the integer form of Moy–Prasad depth (m/r for $D^\times$), preservation of depth under Jacquet–Langlands, the finite groups $D^\times/F^\times K^{m+1}$ and the count of representations of level ≤ 2 in the running example ($5 + 4 + 4$, dimensions $1, 1, 1, 1, 2; 4; 6$), Howe–Corwin admissible pairs in the tame case $p \nmid r$ and the relation of level to the conductor of χ, and the remark that levels are unbounded. Bushnell–Henniart added to the bibliography.

Future directions (new Section 15)

Why level two is a natural stopping point; the two statements (B1), (B2) per level and which failures would be obstructions; the level-three setup (H_3 of order 3^{11}, Clifford theory over Q_5/Q_{10}, the certificate format); two ideas toward an inductive mechanism (an orbit-theoretic description in the spirit of Howe's Kirillov theory; inductive freeness via the central subgroups $\hat{A}_m$, which are central in $\hat{H}_m$ for $m \geq 3$ with no hypothesis); what remains for Target C; and the duality construction.

Numbering

The new lemma shifts two numbers: the level-two theorem is now 13.8 and freeness 13.9; the new sections shift the interface section to 16. Labels unchanged; the scripts' README updated.